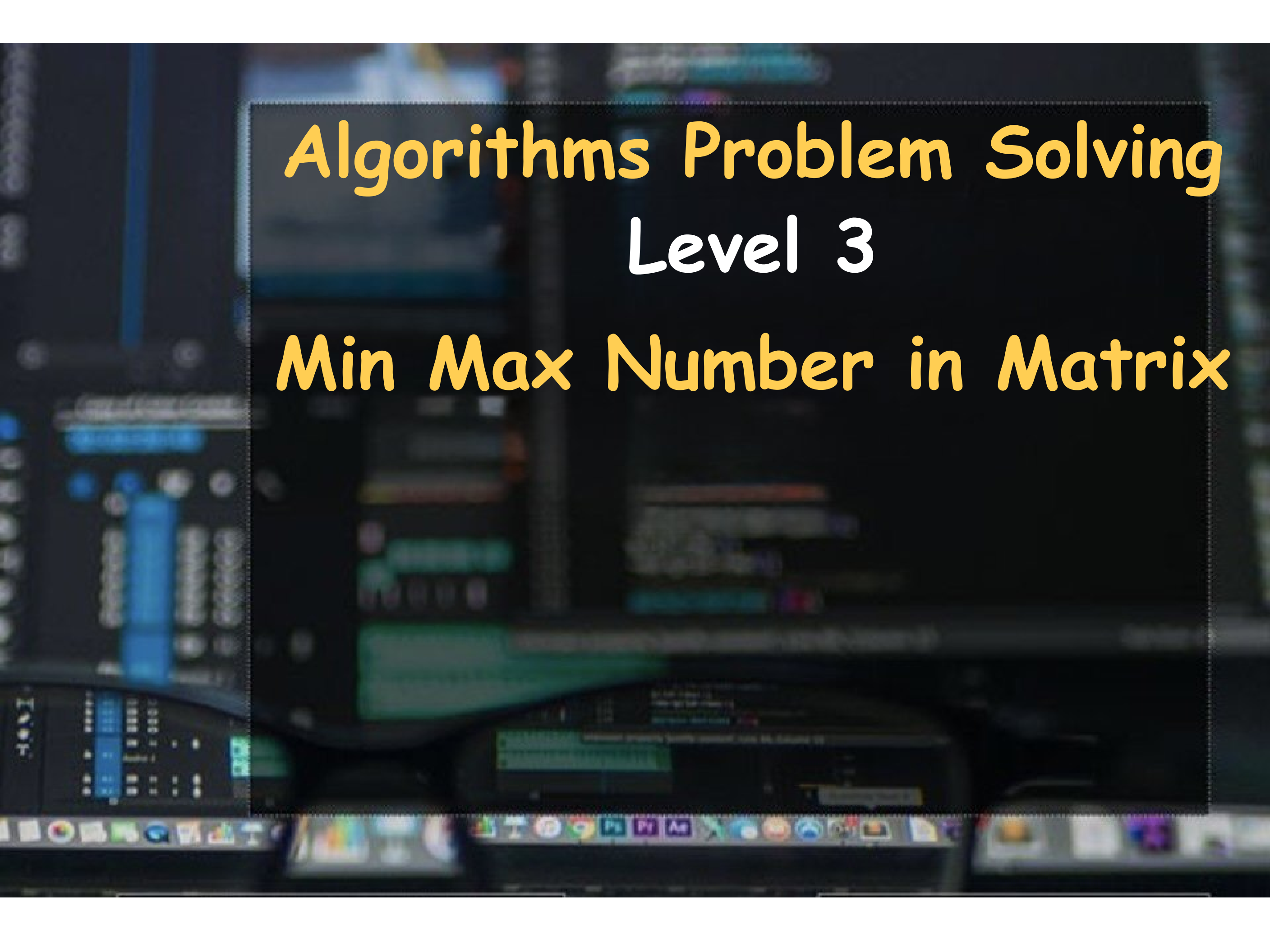
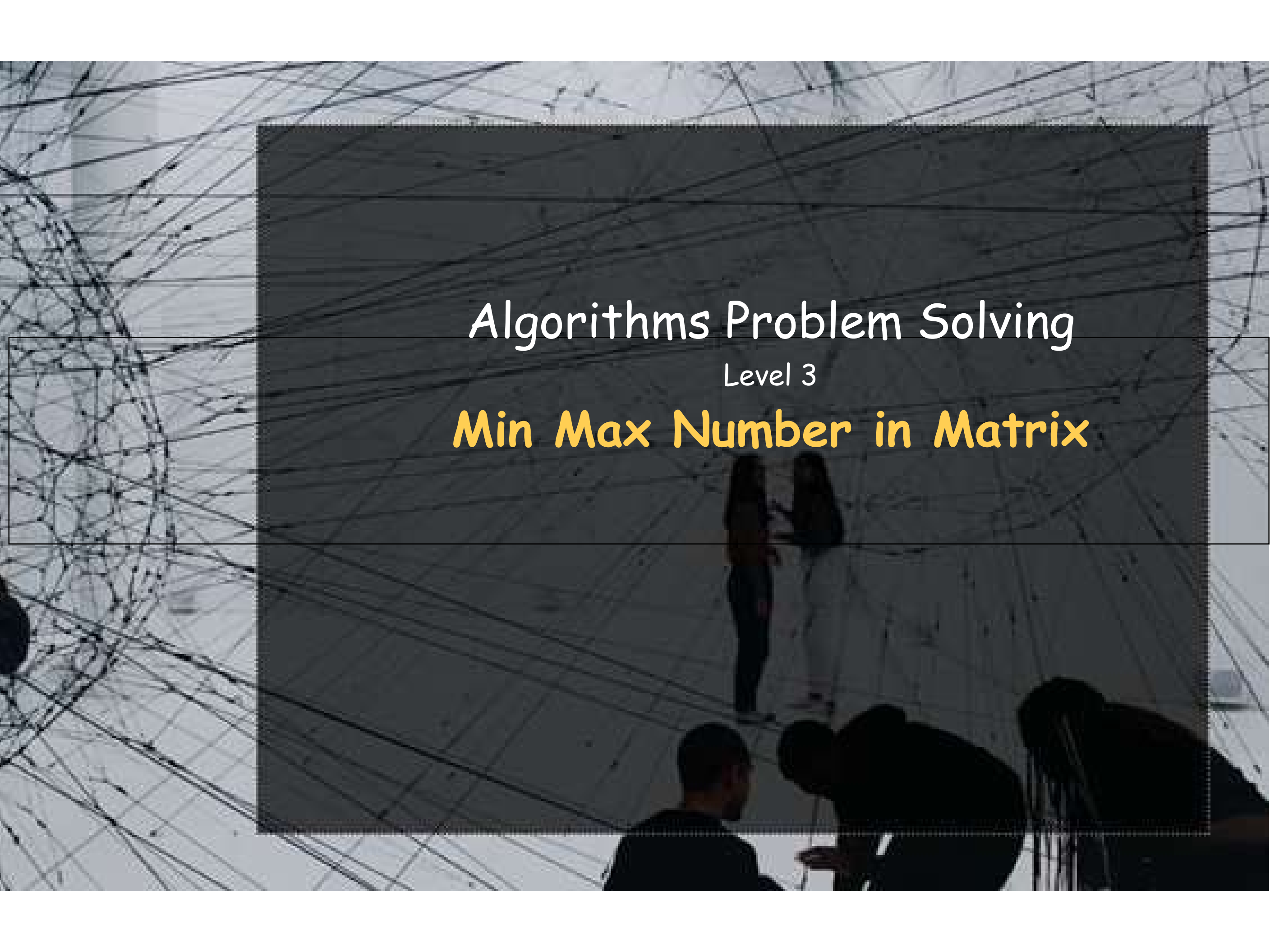




Algorithms Problem Solving Level 3

Min Max Number in Matrix



Algorithms Problem Solving

Level 3

Min Max Number in Matrix

Definition

Extremum Search in 2D Arrays

-  Program scans a matrix to find its smallest and largest elements.

-  The minimum and maximum are determined by comparing every matrix value against the current boundary values.

Explanation

Baseline Initialization

-  Both ``min`` and ``max`` are initialized using ``matrix[0][0]``.
-  This avoids making assumptions about the possible numeric range.

Minimum Tracking

-  Whenever a value smaller than the current ``min`` is found, ``min`` is updated.

Maximum Tracking

-  Whenever a value larger than the current ``max`` is found, ``max`` is updated.

Complete Traversal

-  Every cell must be inspected to guarantee that the true minimum and maximum have been found.

Procedures

1. Fill the matrix.
2. Initialize `min` and `max` with `matrix[0][0]`.
3. Traverse every row and column.
4. Update `min` when a smaller value appears.
5. Update `max` when a larger value appears.
6. Display both results.

💡 Example

Matrix1:

77 5 12

22 20 1

1 0 9

➡ Result

Minimum Number is: 0

Max Number is: 77

Note

- ◆ The algorithm works with positive and negative values.
- ◆ Initializing from the first element makes the algorithm independent of the matrix value range.
- ◆ Time complexity is $O(N \times M)$.

⊘ Warning

- ◆ Do not initialize ``min`` to ``0`` because a matrix containing only positive values would produce an incorrect result.
- ◆ Do not initialize ``max`` to ``0`` when negative values are possible.
- ◆ The matrix must contain at least one element before accessing ``matrix[0][0]``.



Comparison

▨ `MinNumberInMatrix()`

- ✓ Tracks the smallest element.
- ✓ Updates the minimum whenever a smaller value is encountered.
- ✓ Requires a complete matrix scan.

▤ `MaxNumberInMatrix()`

- ✓ Tracks the largest element.
- ✓ Updates the maximum whenever a larger value is encountered.
- ✓ Requires a complete matrix scan.

✦ Summary

- ✓ Initializes boundary values from the first matrix element.
- ✓ Traverses all matrix positions.
- ✓ Dynamically updates minimum and maximum values.
- ✓ Finds the true extrema without assuming a numeric range.

Thank you 😊